CCC 402 –Genetic Engineering-a journey of Life

Genesis of Life

DNA structure

DNA Function,

DNA Replication,

Central Dogma Transcription,

Codons

Translation,

Gene cloning, - 2

PCR,

Mutations

Protein structure, -2

Protein expression, -2

Protein Purification, -2

Importance in Plant and Animal Biotechnology

Quiz 1 Aug 22

Quiz 2 Sep 5

Doubt Clearing

LIST OF LECTURES **Marks Distribution**

Quizes (10 marks each-MCQ)

Final – 40 Questions (~ 15-20 questions short answers – 2 line answers, 2 marks each)

(20 Questions MCQ – 2 marks each)

- **Slides will be uploaded at the end of every week**
- **Reference book is Gene Manipulation Old and Primrose**

(Students are welcome to discuss anything between 1-2 PM, any day)

THE EVOLUTION…

Summary of Darwin's Theory

- Individual organisms in nature differ from one another. Some of this variation is inherited
- Organisms in nature produce more offspring than can survive, and many of those that survive do not reproduce.
- Because more organisms are produced than can survive, members of each species must compete for resources.
- Because each organism is unique, each has different advantages and disadvantages in the struggle for existence.

THE EVOLUTION…

Summary of Darwin's Theory

5. Individuals best suited to their environment survive and reproduce most successfully. The characteristics that make them best suited to their environment are passed on to offspring. Individuals whose characteristics are not as well suited to their environment die or leave fewer offspring.
6. Species change over time. Over long periods, natural selection causes changes in the characteristics of a species, such as in size and form. New species arise, and other species disappear.
7. Species alive today have descended with modification from species that lived in the past.
8. All organisms on Earth are united into a single tree of life by common descent.

DNA Functions

- **Stores genetic information to build and maintain a living organism**
- **Copies itself**

HISTORY OF DNA

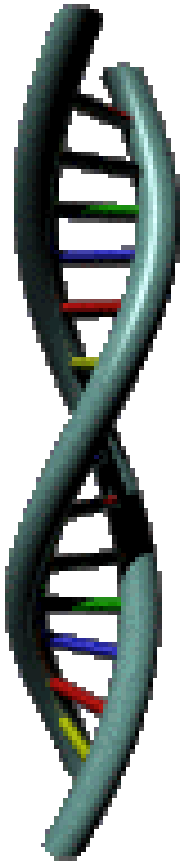
- Discovery of the **DNA double helix**

  A. **Frederick Griffith** – Discovers that a factor in diseased bacteria can transform harmless bacteria into deadly bacteria **(1928)**

  B. **Rosalind Franklin** - X-ray photo of DNA **(1952)**

  C. **Watson and Crick** - described the DNA molecule from Franklin's X-ray **(1953)**

James Watson & Francis Creek 1953



- **Determined the three-dimensional structure of DNA**
- **One of the greatest discoveries in the history of science**
- **Nobel prize in 1962**

Deoxyribonucleic acid (DNA) is a long double-stranded chain of nucleotides

- DNA is the hereditary material passed on from generation to generation.
- DNA is made up of four nucleotides: A, C, G, and T.
- A always pairs with T.
- C always pairs with G.
- The two strands of DNA are in an antiparallel configuration.
- Two complementary DNA strands will separate when heated, and will spontaneously pair together again (hybridize) when cooled.

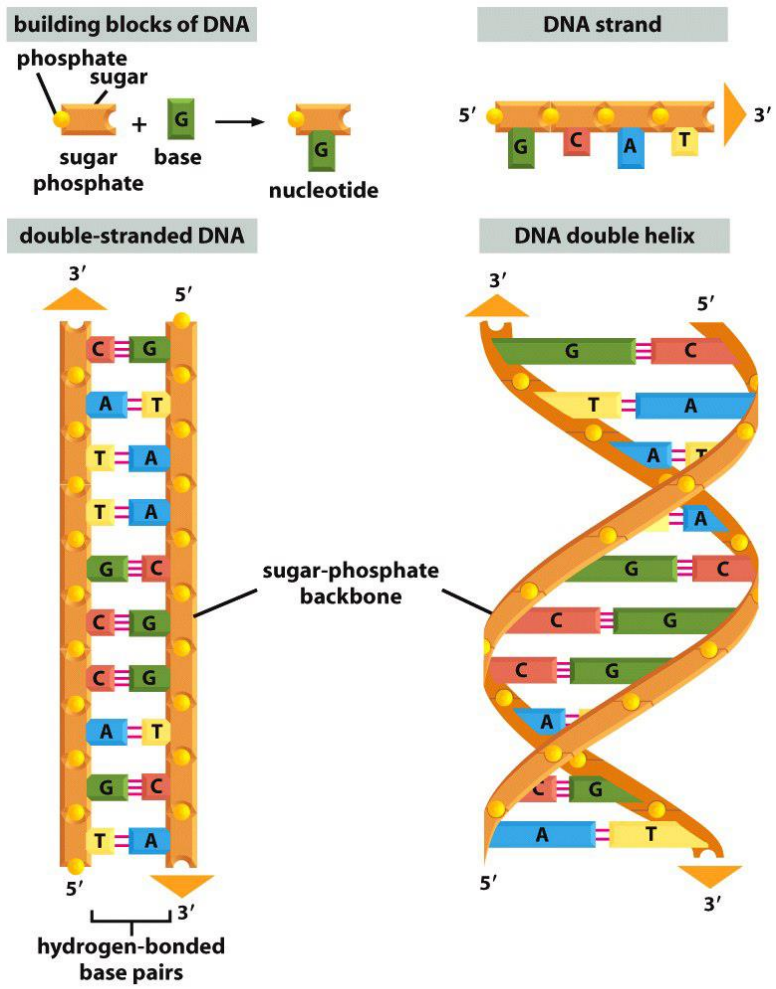
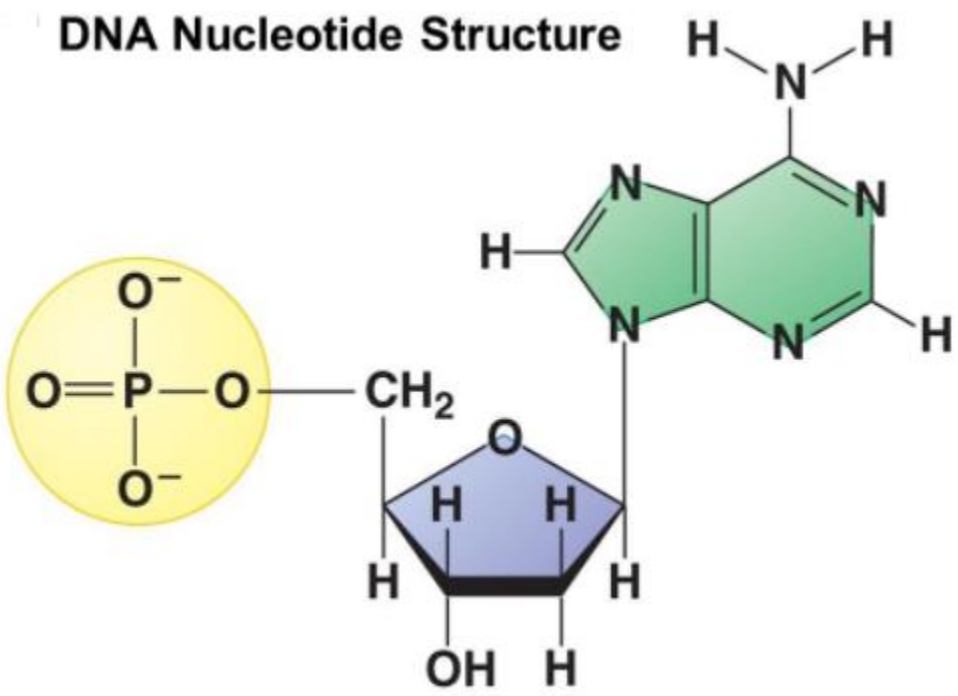
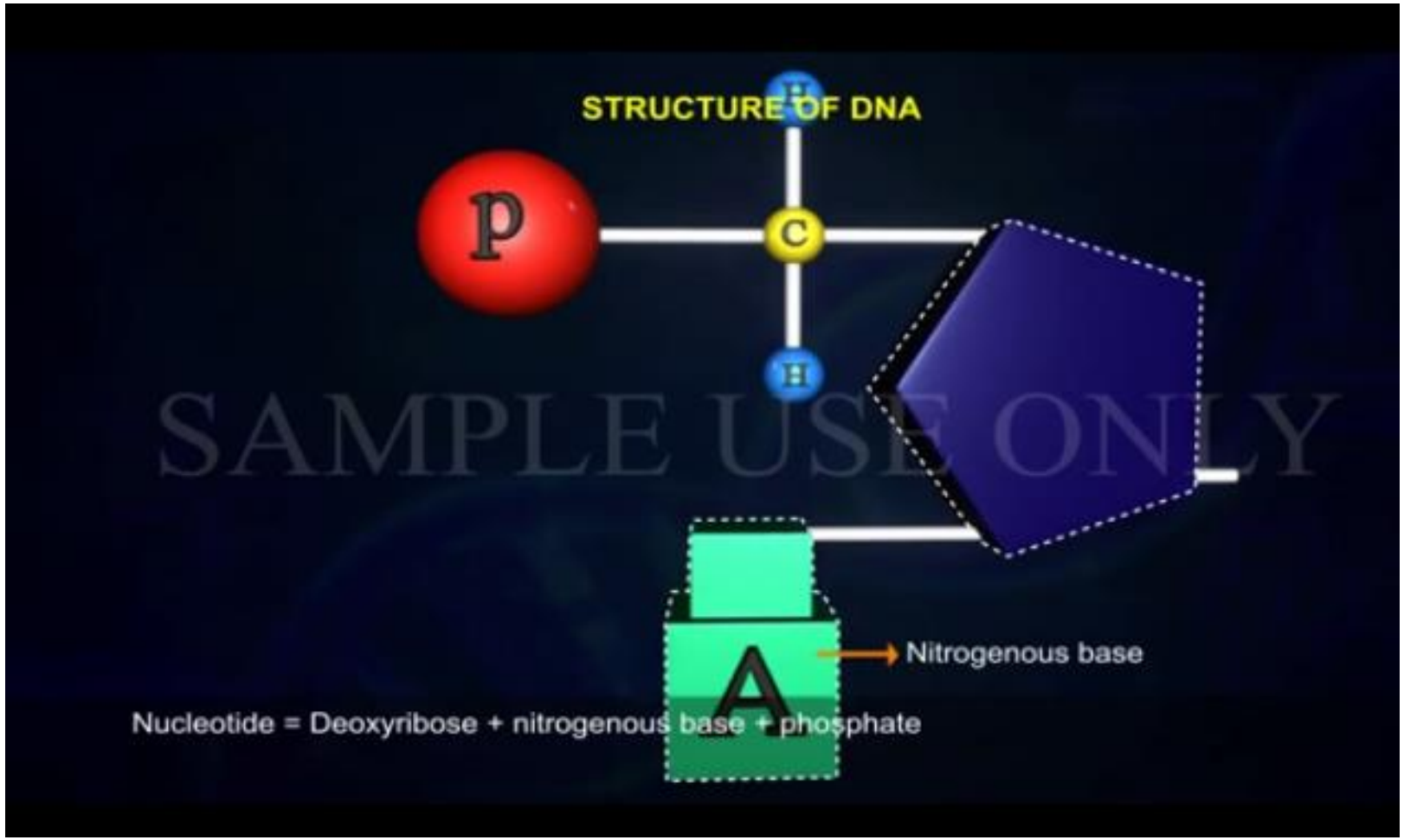


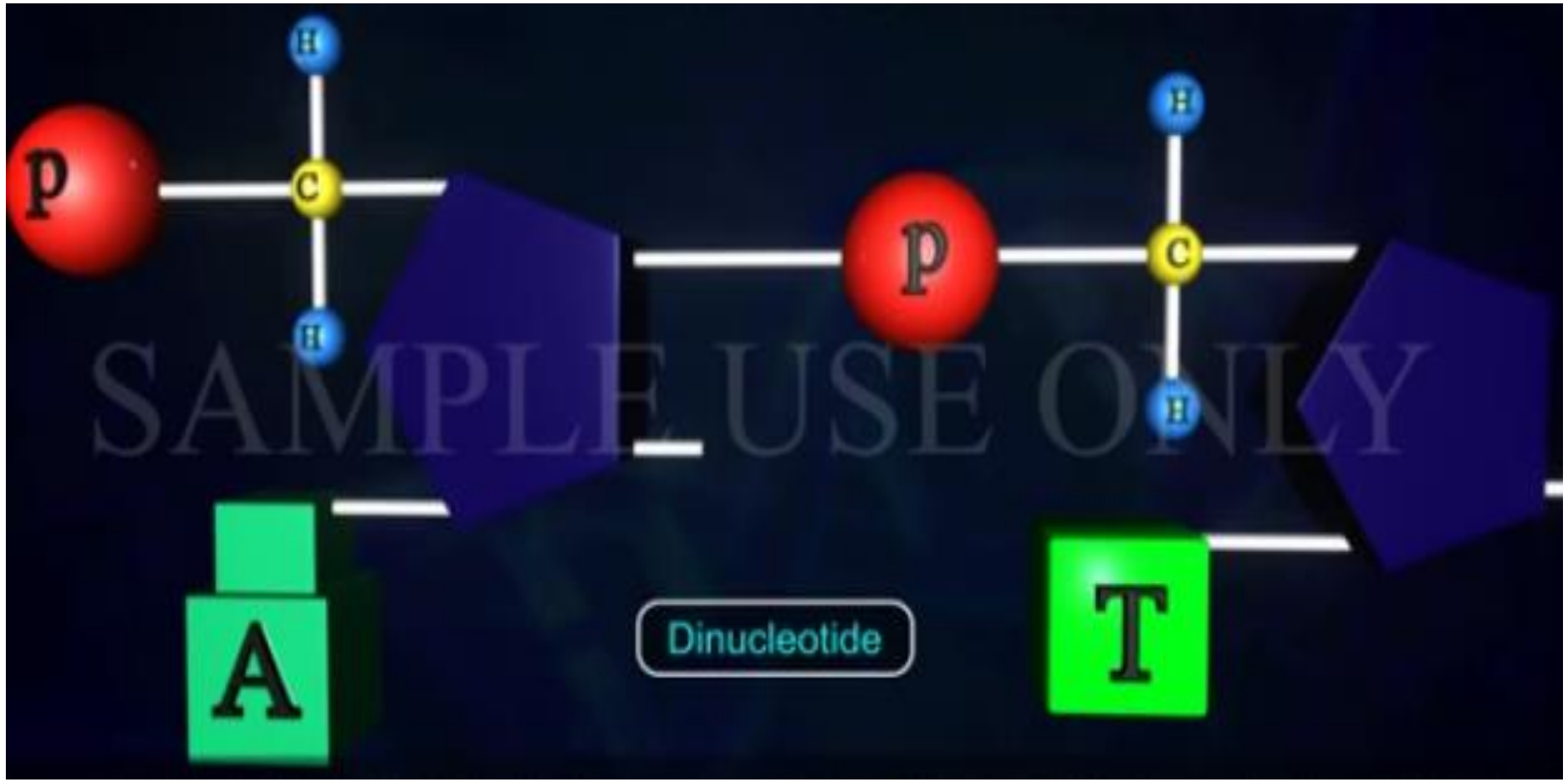
Figure 4-3 Molecular Biology of the Cell 5/e (© Garland Science 2008)


DNA Nucleotide Structure
H
N
H
N
N
H
N
N
H
O⁻
O
P
O
O⁻
CH2
O
H
H
H
H
OH
H

MONUCLEOTIDE

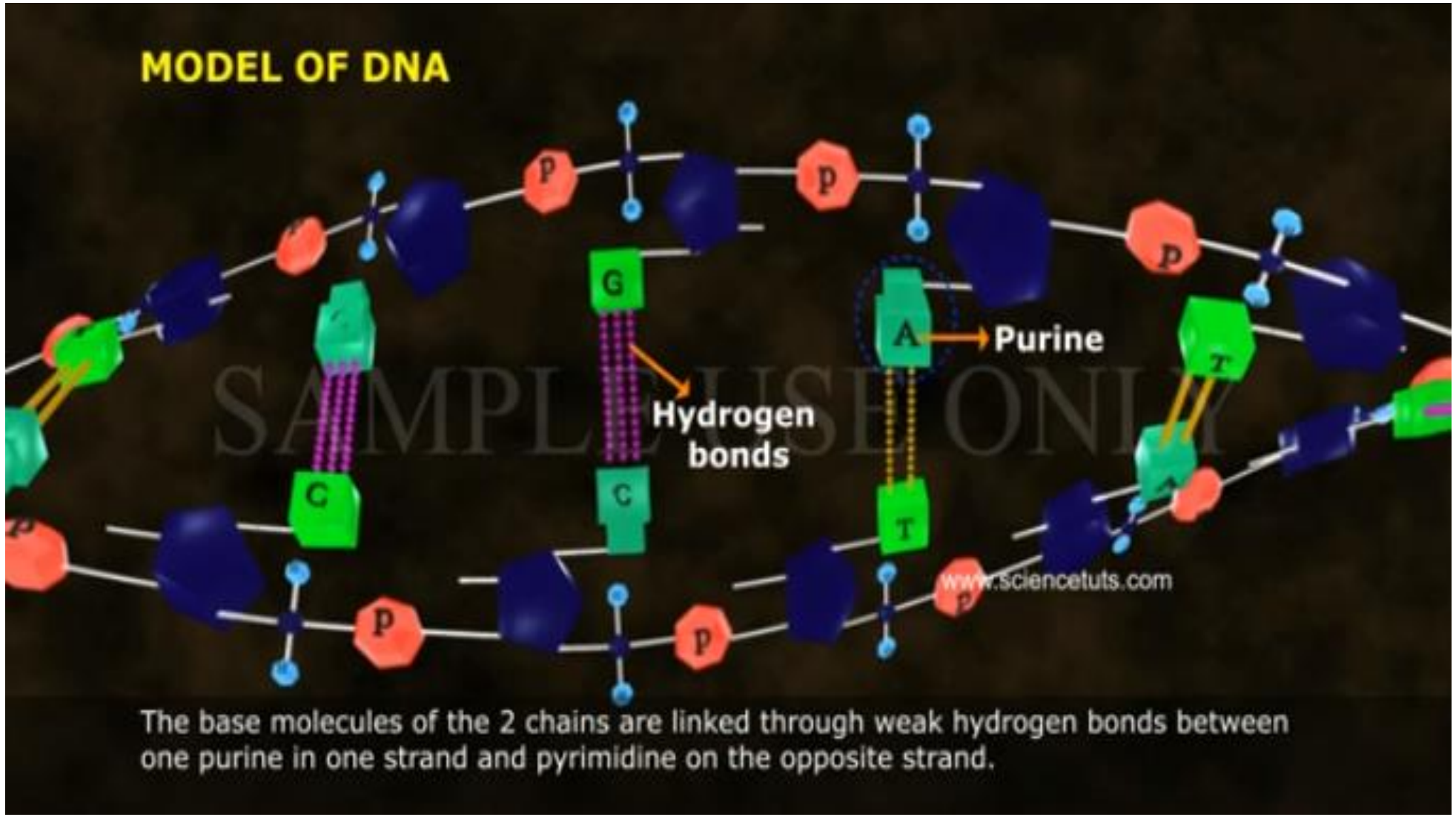


DINUCLEOTIDE

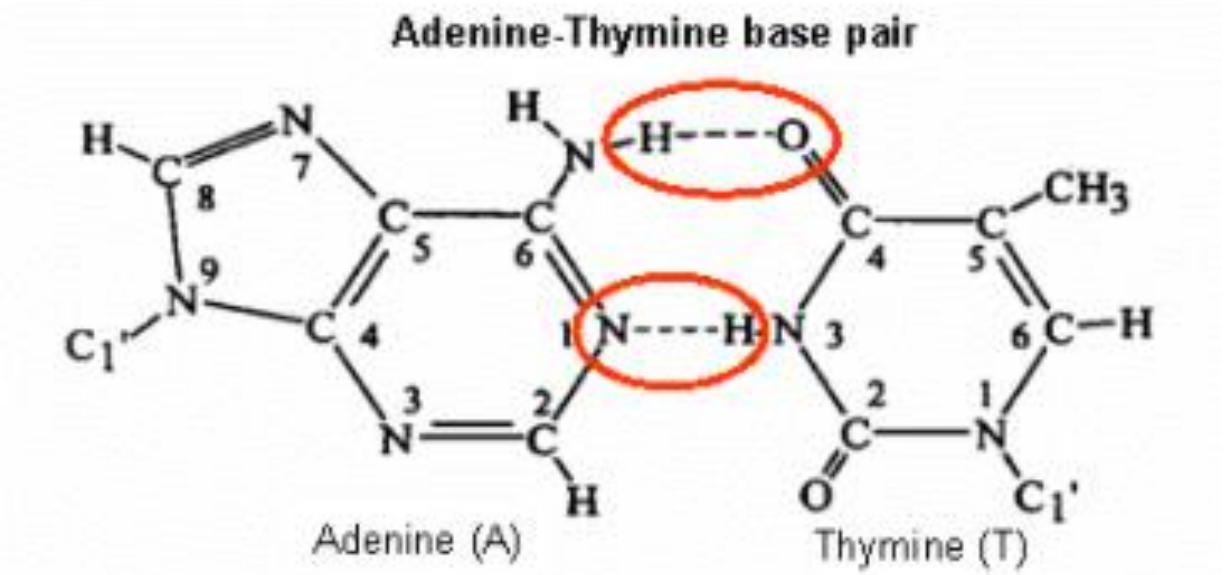


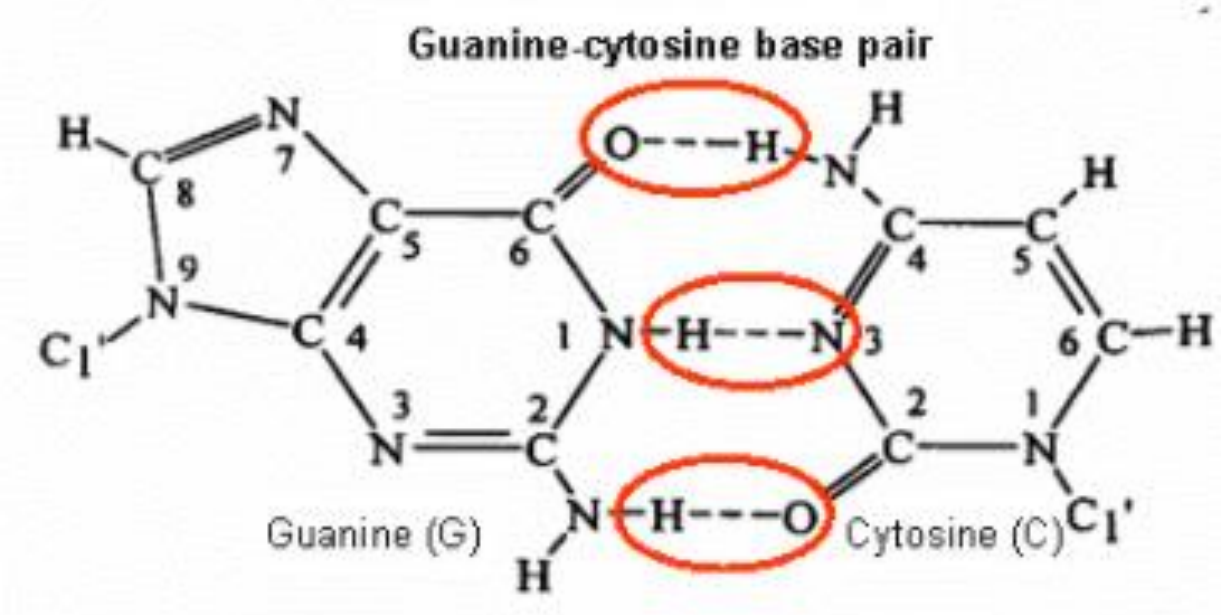
SAMPLE USE ONLY

POLYNUCLEOTIDE

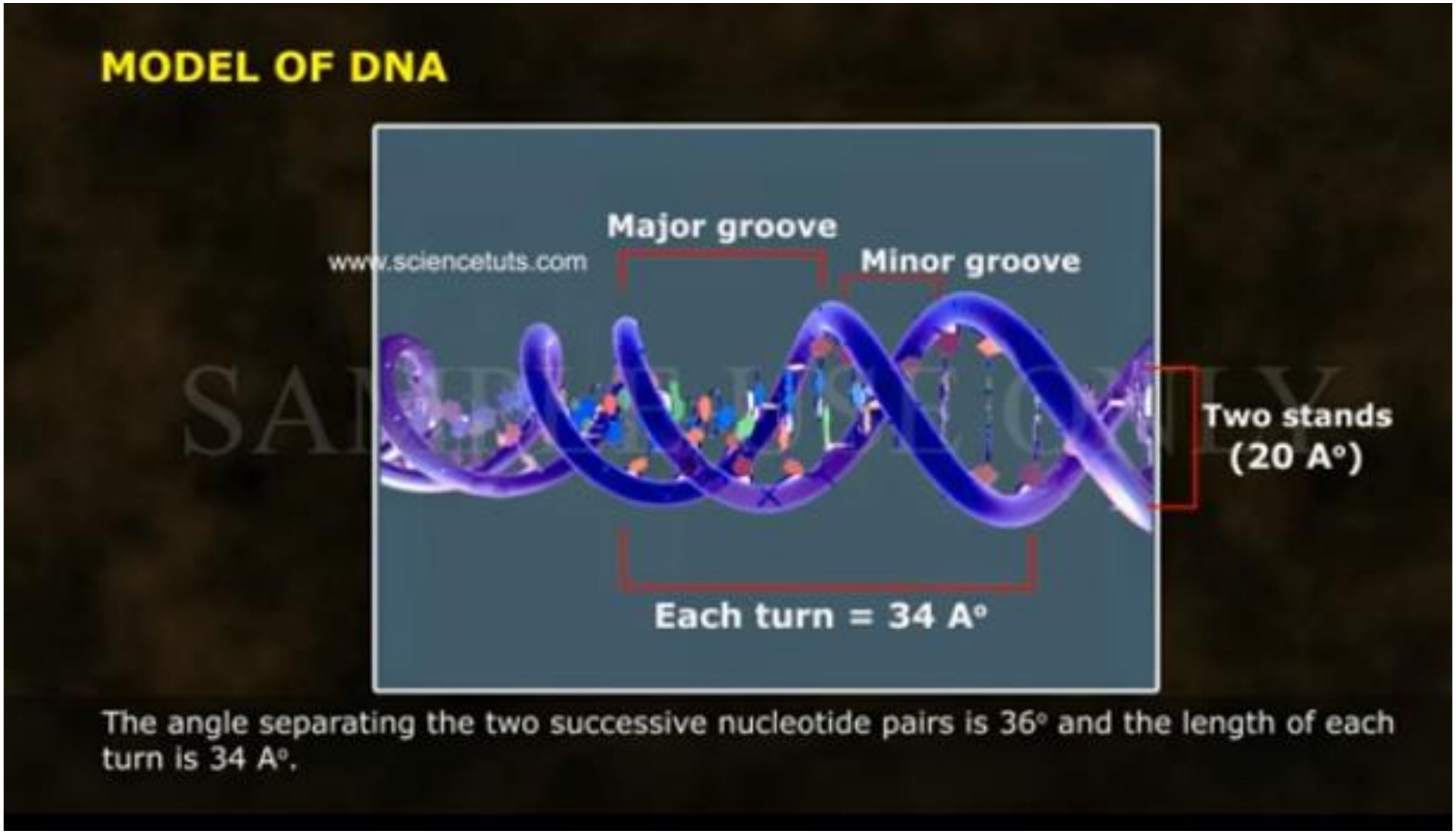


PAIRING BETWEEN NUCLEOTIDES

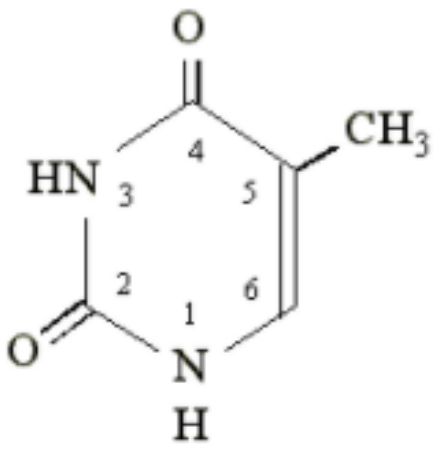




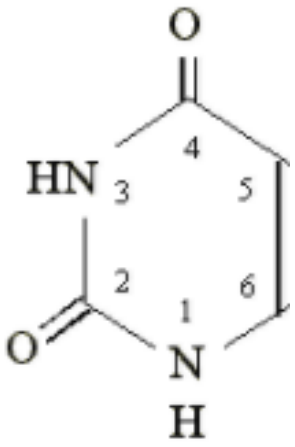
BASE PAIRING IN HELIX



BASES THAT DIFFER IN DNA AND RNA



Thymine
(DNA)



Uracil
(RNA)